

Quantum vs. Classical Correlations in Dijet Hadron Multiplicities: A Finite-Statistics Methodology Note

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Abstract

We develop and validate a complete methodological pipeline for measuring mutual information (MI) between back-to-back jet hadron multiplicities. The naive plug-in entropy estimator is biased upward at finite sample size, in exactly the direction that could mimic a real excess; using a Monte Carlo toy model with known ground truth, we validate two standard corrections (Miller–Madow, shuffle-subtraction) and a Bayesian estimator (NSB), and establish a general error-estimation framework (bootstrap resampling of the full quantity of interest, never its separately-estimated pieces) used throughout everything that follows. The full pipeline, cross-validated between independent Python and ROOT/C++ implementations, is applied to realistic PYTHIA 8 dijet Monte Carlo. Since PYTHIA output carries no independently known true MI, we develop a closure test that calibrates a synthetic model to the data’s own statistics; applied to a full validation sample this finds NSB’s bootstrap uncertainty under-covers by roughly $1.5\text{--}1.7\times$, improving substantially when the same test is run separately within p_T bins. We further identify and correct a distinct systematic: any finite p_T window induces real but kinematically trivial MI, since both jets’ multiplicities and their p_T ’s are correlated through momentum balance. A mixed-event subtraction – pairing each jet with a p_T -matched partner from a different event – isolates the beyond-kinematics MI; closure tests, both on synthetic data and at full statistics on the real sample’s own fitted kinematics, show the naive estimator’s finite-sample bias nearly cancels between the same-event and mixed-event terms, so the raw difference is close to unbiased even before rigorous correction. Applied to the real validation sample, the subtraction finds a nonzero beyond-kinematics component of order 0.003 nats – consistent across all four estimator methods and plausibly attributable to shared underlying-event activity rather than anything exotic. Throughout, we are careful about what this measurement can and cannot establish: a rigorously measured $I(A:B)$, with its estimator bias removed, its uncertainty calibrated, and its trivial kinematic component separated out, is the necessary foundation for any later physics interpretation, but not by itself a quantum witness. The result is a validated, closure-tested toolkit – estimator, calibration factors, and kinematic-background subtraction – ready to apply to real STAR/ALICE dijet data.

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1 Physics motivation

Take a back-to-back dijet event (e.g. from e^+e^- , DIS, or pp collisions at RHIC/STAR). Treat jet A and jet B as the two subsystems of a bipartite system whose “state” is accessed only through observed hadron multiplicities, via **local parton–hadron duality (LPHD)**: the hadron multiplicity distribution stands in for the underlying parton Fock-state distribution.

The Schmidt-decomposition ansatz (Kharzeev–Levin; Peschanski–Seki): if each Fock/parton-number sector n carries probability $P(n)$, and the reduced density matrix of a jet is diagonal in n , then the entanglement entropy of that jet reduces to the **Shannon entropy of its hadron multiplicity distribution**,

$$S_A = - \sum_n P_A(n) \ln P_A(n), \quad (1)$$

and likewise for S_B . This turns a genuinely quantum quantity (entanglement entropy) into something directly measurable from ordinary multiplicity counting – the whole program rests on this ansatz.

The physics question: by additionally measuring the *joint* multiplicity distribution $P_{AB}(n_A, n_B)$ across both jets in the same event, we can construct the mutual information $I(A:B)$ between the two jets (defined in Sec. 2), a measure of how strongly correlated their multiplicities are. Under the Schmidt-decomposition ansatz above, a nonzero $I(A:B)$ is at least consistent with correlations surviving from the partonic state through hadronization – possibly including a genuine quantum (entanglement) contribution, on top of ordinary classical correlation (shared color flow, momentum conservation, or other QCD-driven but fundamentally classical dynamics).

Why this note does not chase a rigorous witness claim. A natural sharper test would be conditional-entropy negativity, $S(B|A) = S_{AB} - S_A < 0$ (equivalently $I(A:B) > \min(S_A, S_B)$) – classically impossible, since Shannon conditional entropy of *any* probability distribution built from real, counted outcomes is mathematically guaranteed to be ≥ 0 . That is not a physical assumption but a theorem about frequencies of definite measurement outcomes, and it cuts both ways: it would make a genuine violation unambiguous *if* the Schmidt-decomposition ansatz literally held all the way through hadronization, but it also means an *observed* violation in real, finite-sample data can only ever be a statistical/estimator artifact – never directly-observed physics – since the true classical value is provably non-negative by construction, regardless of the underlying dynamics. Separately, genuine (Ollivier–Zurek) quantum discord, the model-independent notion of “quantumness” in a correlation, is identically zero for any classical joint distribution built from count histograms; extracting a nonzero discord needs actual quantum-state tomography (e.g. the parallel $\Lambda\bar{\Lambda}$ spin-density-matrix program), not multiplicity counting. For both reasons, this note does not treat $I(A:B)$ measured from hadron multiplicities as a stand-alone quantum witness. What it *is* good for is measuring $I(A:B)$ itself rigorously and reproducibly – correcting its finite-sample bias, calibrating its uncertainty, and separating genuine dynamical correlation from trivial kinematic correlation – which is a well-posed statistical problem in its own right, and the necessary foundation before any comparison to the Schmidt-decomposition prediction can be taken seriously. That is the entire scope of what follows.

Why this is subtle in practice: any such measurement is only as good as correctly accounting for the finite size of real data samples, since MI estimated from a finite event sample is **always statistically biased**, and – as we show below – biased in exactly the direction that could be mistaken for a large excess if not handled carefully. This note works through that problem end to end: bias, correction, uncertainty, closure testing, and a further kinematic background that mimics real correlation for entirely mundane reasons.

2 The observable: mutual information

From the per-jet and joint hadron-multiplicity distributions we build three Shannon entropies and combine them into the mutual information between the two jets:

$$S_A, \quad S_B, \quad S_{AB} \quad \Longrightarrow \quad I(A:B) = S_A + S_B - S_{AB}, \quad (2)$$

where $S_A = -\sum_n P_A(n) \ln P_A(n)$ is the Shannon entropy of jet A 's hadron-multiplicity distribution (likewise S_B), and $S_{AB} = -\sum_{n_A, n_B} P_{AB}(n_A, n_B) \ln P_{AB}(n_A, n_B)$ is the joint entropy of the two jets' multiplicities measured together in the same event. $I(A:B) \geq 0$ always, with equality only if N_A and N_B are statistically independent; it is the quantity measured, corrected, and calibrated throughout the rest of this note. As discussed in Sec. 1, $I(A:B) > 0$ on its own only establishes that the two jets' multiplicities are correlated *somehow* – consistent with, but not proof of, the entanglement-entropy interpretation motivating this measurement.

3 The statistical problem: finite-sample bias

Every quantity above is estimated from a finite event sample, and the naive (“plug-in”) entropy estimator has a **systematic, structural bias** that turns out to point in exactly the dangerous direction for this measurement.

Where the bias comes from. The naive estimator replaces true probabilities p_i with empirical frequencies $\hat{p}_i = n_i/N$,

$$\hat{S} = -\sum_i \hat{p}_i \ln \hat{p}_i. \quad (3)$$

Because $x \ln x$ is convex (equivalently $-x \ln x$ is concave), Jensen's inequality guarantees

$$\langle \hat{S} \rangle < S_{\text{true}} \quad \text{for any finite } N, \quad (4)$$

i.e. the plug-in estimator always underestimates entropy. This isn't a numerical artifact; with only N draws one can never fully resolve a distribution's tails, and that missing resolution always looks like *less* apparent randomness than there truly is.

Leading-order size of the effect (Miller 1955; Madow), for a distribution over K possible outcomes:

$$\langle \hat{S} \rangle \approx S_{\text{true}} - \frac{K-1}{2N}. \quad (5)$$

Fewer events (N small) or finer binning (K large) both make the downward bias worse.

Why this flips sign and biases $I(A:B)$ upward. Combining the individual biases of $\hat{S}_A, \hat{S}_B, \hat{S}_{AB}$ with correct signs,

$$\langle \hat{I} \rangle \approx I_{\text{true}} + \frac{(K_{AB}-1) - (K_A-1) - (K_B-1)}{2N} = I_{\text{true}} + \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (6)$$

The joint distribution lives in a much larger space than either marginal alone, so $\hat{S}_{AB}$ suffers a *much larger* downward bias than $\hat{S}_A + \hat{S}_B$ does – and since S_{AB} enters with a **minus** sign, its extra negative bias flips into a **positive** bias on $\hat{I}$. In the extreme case $N \ll K_A K_B$ (badly undersampled joint histogram), this produces large *phantom* apparent mutual information even

for truly independent variables – sampling noise alone can masquerade as a real correlation if not corrected.

A commonly-quoted shortcut version of Eq. (6) is $(K_A - 1)(K_B - 1)/(2N)$; this is not a different formula, it is Eq. (6) evaluated under the extra assumption $K_{AB} = K_A K_B$, i.e. a fully and uniformly occupied joint space. That assumption only holds for independent variables; for correlated data the observed K_{AB} is much smaller than $K_A K_B$ (the joint distribution concentrates on a narrow ridge rather than filling the full grid), so this shortcut badly *overestimates* the bias – a pitfall we hit directly in Sec. 5.

Practical consequence. Comparing the same underlying physics across data samples of different size – different STAR running periods, different luminosity, different jet-cone or multiplicity binning choices – will report systematically different central MI values purely from this effect, with zero change in the actual QCD physics. Secs. 4–6 demonstrate this concretely with a toy model and work through three ways to correct for it.

4 Toy Monte Carlo test bed

To demonstrate the bias concretely and validate corrections against a known ground truth, we use a controlled toy model: two back-to-back jets, A and B , with a shared latent hard-process fluctuation Λ (e.g. fluctuating parton virtuality / initial-state energy scale) drawn from a Gamma distribution. Conditional on Λ , each jet’s hadron multiplicity is Poisson distributed,

$$\Lambda \sim \text{Gamma}(k, \theta), \quad N_A | \Lambda \sim \text{Poisson}(\Lambda), \quad N_B | \Lambda \sim \text{Poisson}(c\Lambda). \quad (7)$$

Marginalizing over Λ gives each jet a **Negative Binomial** multiplicity distribution (the standard KNO-like form used for hadron multiplicities), and the shared Λ induces a genuine – but purely **classical** – correlation between N_A and N_B , i.e. $I_{\text{true}} > 0$. This is a convenient sandbox: we know the true joint distribution (to arbitrary precision, from a huge reference sample), so we can measure the *bias* of any estimator directly against a known answer. Parameters used throughout: $k = 3.0$, $\theta = 8.0$, $c = 0.9$ (giving $\text{corr}(N_A, N_B) \approx 0.88$).

4.1 Demonstration: two samples, same true distribution, different size

We generate a “small” event sample ($N = 500$; e.g. a single limited run / low-luminosity STAR-like dataset), a “large” event sample ($N = 200,000$; high-statistics dataset), and a very large reference sample ($N = 5 \times 10^6$) to pin down I_{true} – all drawn from the *exact same underlying physics*.

Table 1: Naive $\hat{I}$ for three samples of the same true physics.

Sample	N	K_A	K_B	K_{AB}	$\hat{I}_{\text{naive}}$ (nats)
small	500	67	60	390	1.7424
large	200,000	129	113	3949	0.7463
reference	5,000,000	159	147	7022	0.7329 ($\approx I_{\text{true}}$)

We see $\hat{I}_{\text{small}} > \hat{I}_{\text{large}} \approx \hat{I}_{\text{ref}} \equiv I_{\text{true}}$ – **the same physics, analyzed with fewer events, reports a systematically larger MI**. This is the bias predicted in Sec. 3, not a change in physics.

The bias comes entirely from how well-populated the joint histogram is. Figure 1 shows (top) the marginal hadron-multiplicity distributions $P_A(n)$, overlaid for the three samples – these look reasonably similar even at $N = 500$ because there are only $O(10^2)$ bins to fill – and (bottom) the 2D joint histograms $P_{AB}(n_A, n_B)$ on the same color scale, where the difference is dramatic: at $N = 500$ the joint plane is almost empty, while at $N = 5 \times 10^6$ it is a smooth, fully-resolved correlated ridge. A sparse, “salt-and-pepper” joint histogram is exactly what drives $\hat{S}_{AB}$ artificially low, and hence $\hat{I}$ artificially high.

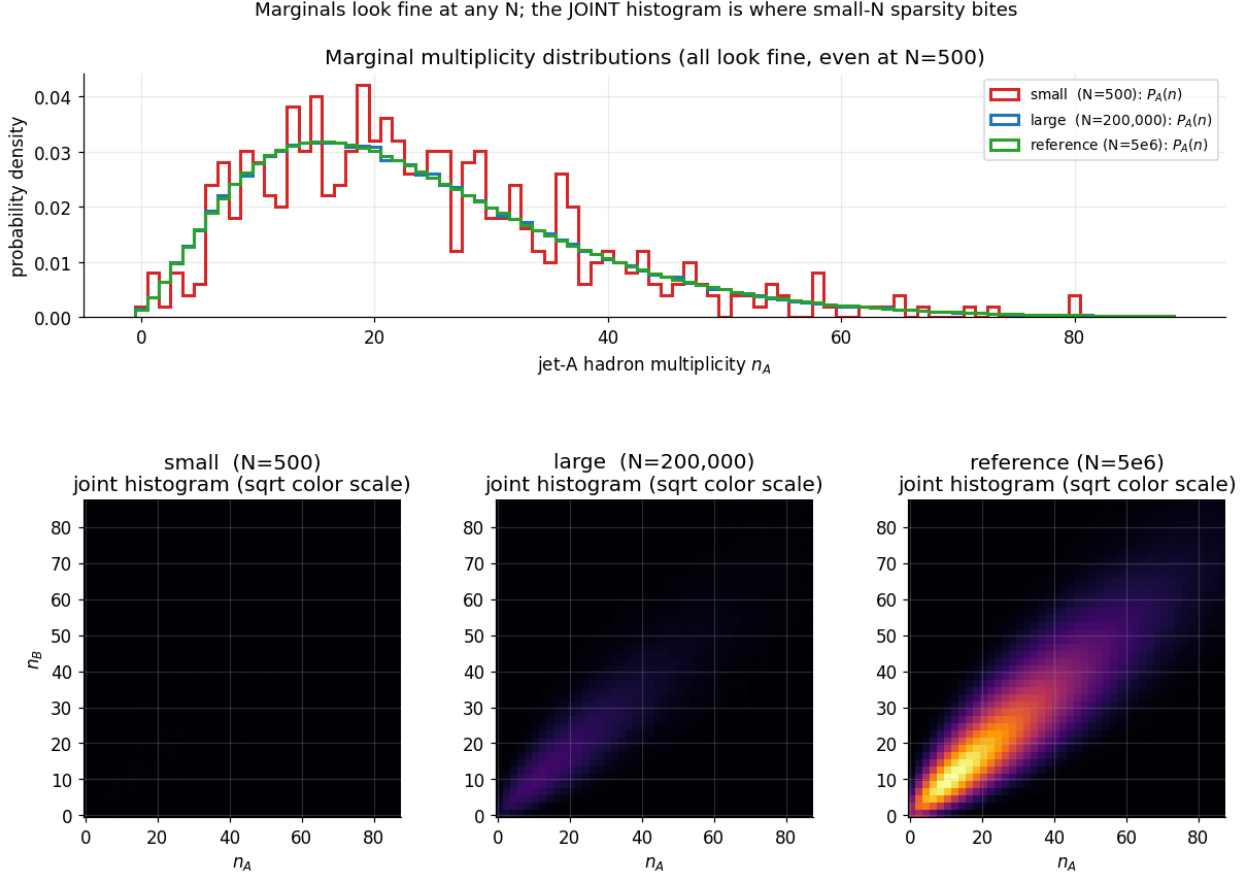


Figure 1: Marginal multiplicity distributions (top) and 2D joint histograms (bottom, square-root color scale) for the small, large, and reference samples. Marginals look consistent at any N ; the joint histogram is where small- N sparsity bites.

4.2 Bias vs. sample size

Repeating the naive estimate over a range of N (with several repetitions per N to see the scatter) and plotting $\hat{I}_{\text{naive}}$ against $1/N$ (Fig. 2) shows the systematic downward drift (bias shrinking) as N grows, converging toward the flat reference line, and the expected near-linear behavior in $1/N$: the intercept of that line is a cheap, model-independent estimate of the bias-corrected/true MI. The extrapolated intercept, 0.9427 nats, lands reasonably close to the reference value of 0.7329 nats.

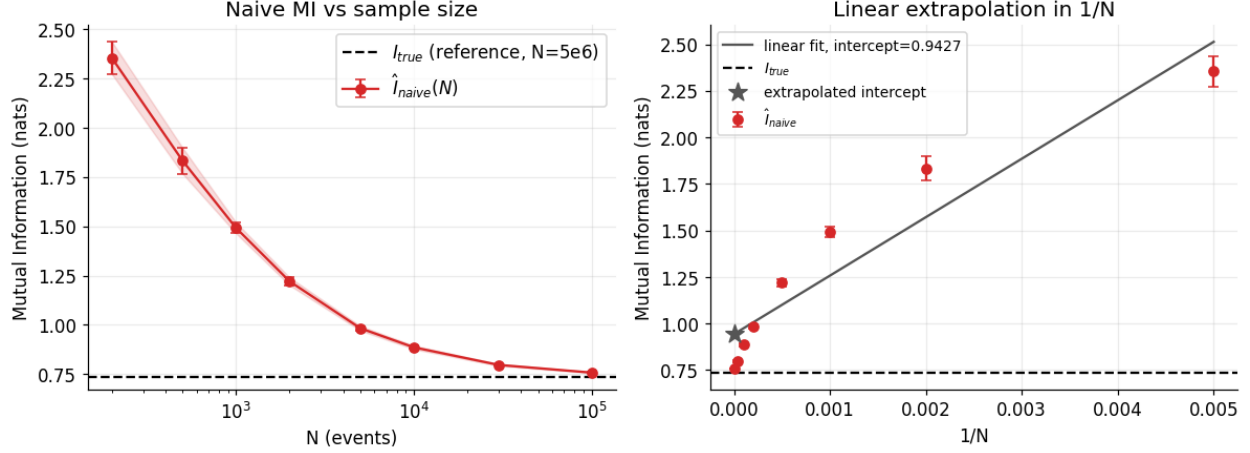


Figure 2: Left: naive $\hat{I}$ vs. sample size N , converging to the reference value. Right: the same data plotted against $1/N$, with a linear extrapolation to $1/N \rightarrow 0$.

5 Corrections: Miller–Madow and shuffle-subtraction

5.1 Miller–Madow correction

The exact leading-order Miller–Madow correction applies the $-(K-1)/(2N)$ term to *each* of S_A, S_B, S_{AB} separately and combines them:

$$\hat{I}_{MM} = \hat{I}_{\text{naive}} - \frac{(K_{AB} - 1) - (K_A - 1) - (K_B - 1)}{2N} = \hat{I}_{\text{naive}} - \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (8)$$

Crucially, K_{AB} here must be the *actually observed* number of occupied joint bins, not the product $K_A K_B$: for correlated variables (our case, by construction) the joint distribution concentrates on a narrow ridge and $K_{AB} \ll K_A K_B$, so using the product in place of the observed value wildly overcorrects and can drive $\hat{I}_{MM}$ negative.

5.2 Shuffle-subtraction (label-permutation null)

Randomly permuting the B -jet labels relative to the A -jet labels *within the same sample* destroys any true correlation ($I_{\text{true, shuffled}} = 0$ by construction) while reproducing the same finite-sample bias. Averaging over many shuffles gives a direct empirical estimate of the bias to subtract:

$$\hat{I}_{\text{corrected}} = \hat{I}_{\text{naive}} - \langle \hat{I}_{\text{shuffled}} \rangle. \quad (9)$$

This makes no asymptotic assumption and remains reliable even when $N \lesssim K_A K_B$, where Miller–Madow starts to break down. It has its own bias, however: shuffling spreads the joint distribution over *more* bins than the true correlated data occupies, so the shuffled surrogate is *more* under-sampled than the real data at the same N . Shuffle-subtraction therefore tends to systematically **overestimate** the bias and **overcorrect**, pulling $\hat{I}_{\text{corrected}}$ below I_{true} when the true correlation is strong.

With the correct Miller–Madow formula (using the observed K_{AB}), both corrections pull the small- N estimates much closer to I_{true} than the naive value and stay well-behaved (no more diving negative). Neither lands exactly on I_{true} at very small N : Miller–Madow is only a first-order

Table 2: Naive, Miller–Madow, and shuffle-corrected $\hat{I}$ vs. N .

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	I_{true} (ref.)
200	2.2764	2.0889	0.0657	0.7329
500	1.8694	1.6014	0.1436	0.7329
1000	1.4271	1.1756	0.2422	0.7329
2000	1.1873	0.9866	0.3809	0.7329
5000	0.9766	0.8539	0.5592	0.7329
20000	0.8259	0.7750	0.6792	0.7329

truncation of the true bias expansion, and shuffle-subtraction overcorrects when the true correlation is strong, as explained above. Both effects shrink as N grows.

The shuffle procedure gives more than a bias correction – the spread of $\hat{I}_{\text{shuffle}}$ over many permutations *is* the null distribution for “ $I_{\text{true}} = 0$ ” at that N and binning (Fig. 3). Plotting $\hat{I}_{\text{naive}}$ against that null distribution is a direct visual significance check: at $N = 500$ the null distribution itself is already centered well above zero (pure bias) and the naive value sits close to or inside this null cloud, so on its own it would not be significant evidence of correlation, even though the jets are truly correlated by construction. By $N = 20,000$ the null has collapsed toward zero and the naive value clearly separates from it.

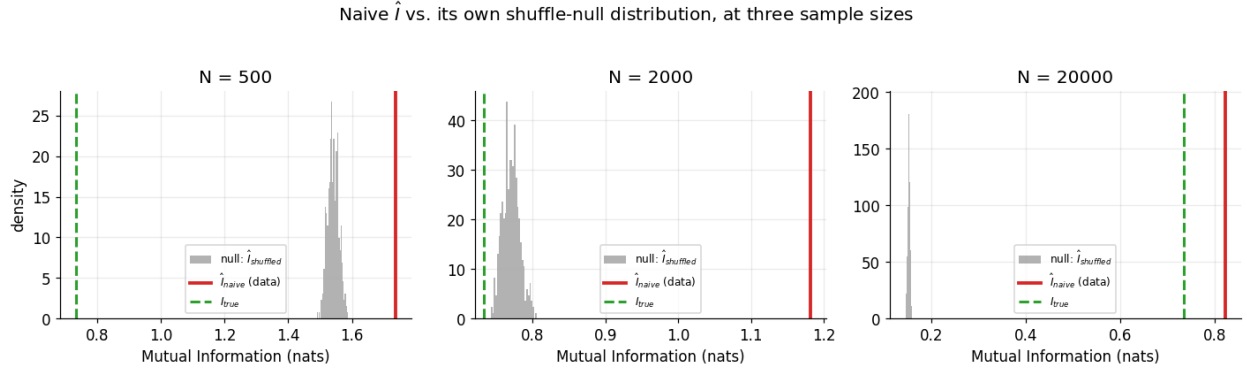


Figure 3: Naive $\hat{I}$ (vertical line) against its own shuffle-null distribution (histogram), at three sample sizes. Dashed line: true value.

Figure 4 summarizes naive vs. corrected MI across sample sizes, and Fig. 5 plots $|\hat{I} - I_{\text{true}}|$ on a log-log scale, which is the honest way to compare convergence rates. Both corrections sit visibly below the naive curve at fixed N ; where the corrected residual curves run roughly parallel to the naive curve rather than dropping faster, that indicates the residual error is increasingly dominated by sub-leading bias terms that neither method captures (Miller–Madow: higher-order $1/N^2, \dots$ terms; shuffle: the systematic overcorrection discussed above) – exactly the gap the more rigorous fix in Sec. 6 is designed to close.

6 The rigorous fix: NSB Bayesian entropy estimator

Both corrections above are still, fundamentally, **point corrections derived under a large- N assumption** – they estimate one number (the bias) and subtract it. The **Nemenman–Shafee–**

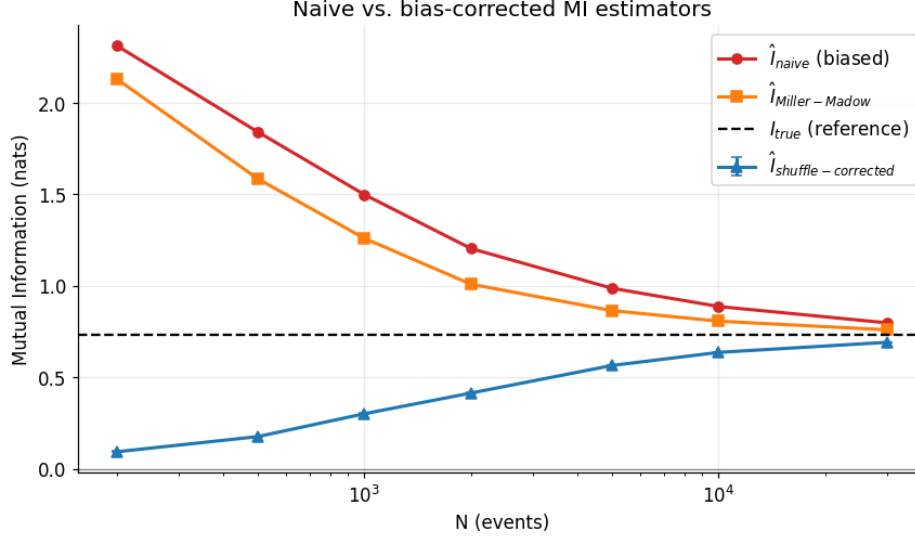


Figure 4: Naive vs. bias-corrected MI estimators across sample sizes.

Bialek (NSB) estimator takes a genuinely different approach: instead of correcting a biased point estimate, it computes the **Bayesian posterior mean entropy** directly, using a prior specifically constructed to avoid smuggling in any implicit assumption about how spread-out the distribution is. This is the standard tool for the severely-undersampled regime ($N \lesssim K$) in the neuroscience spike-train MI literature – precisely our regime once hadron multiplicities are binned in 2D.

The construction, in four steps:

1. Put a symmetric Dirichlet prior $P(p|\beta) \propto \prod_i p_i^{\beta-1}$ over the K possible bin probabilities. For any fixed β , the posterior mean entropy given counts $\{n_i\}$ has a closed form,

$$\langle S \rangle_{\beta,n} = \psi_0(N + K\beta + 1) - \sum_i \frac{n_i + \beta}{N + K\beta} \psi_0(n_i + \beta + 1). \quad (10)$$

2. **The catch:** a flat prior on β does *not* give a flat prior on entropy – for large K , $P(S|\beta)$ is sharply peaked as a function of β . Fixing any single β (as ordinary add-one/Laplace/Jeffreys smoothing does) silently assumes an entropy value before looking at the data.
3. NSB’s fix: reweight β by the hyperprior that makes the *induced* prior on S flat – literally the rate of change of the prior mean entropy with β ,

$$\rho(\beta) = K \psi_1(K\beta + 1) - \psi_1(\beta + 1). \quad (11)$$

4. Average over β , weighted by both this hyperprior and the Dirichlet–multinomial evidence $P(n|\beta)$,

$$\hat{S}_{\text{NSB}} = \frac{\int_0^\infty d\beta \rho(\beta) P(n|\beta) \langle S \rangle_{\beta,n}}{\int_0^\infty d\beta \rho(\beta) P(n|\beta)}. \quad (12)$$

This reduces to a tractable 1D numerical integral. We apply it separately to S_A, S_B, S_{AB} (each needs its own alphabet size K : for the joint distribution, $K_{AB}^{\text{alphabet}} = K_A^{\text{alphabet}} \times K_B^{\text{alphabet}}$, the full product space of possible outcomes – not the observed K_{AB} count from before, since NSB needs

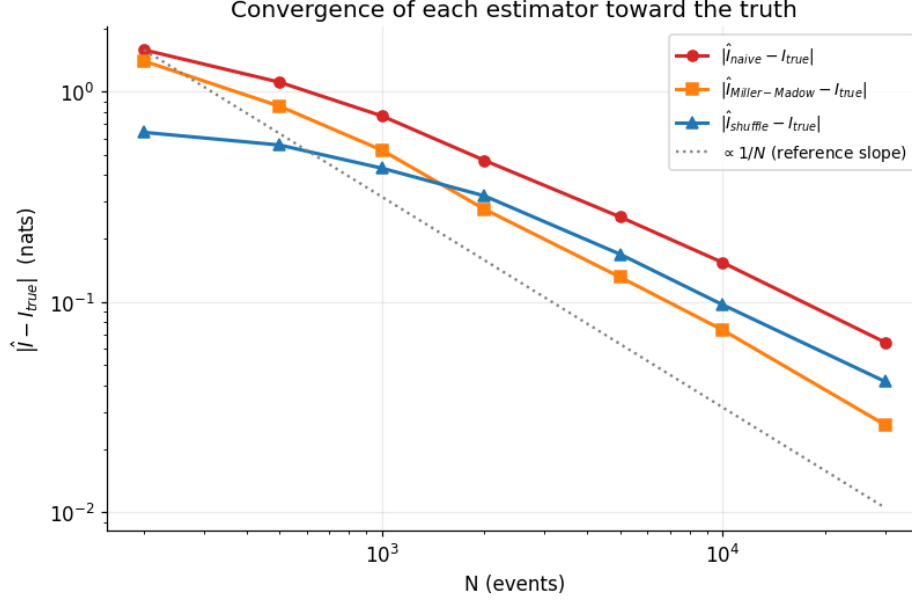


Figure 5: Convergence of each estimator toward the true value, on a log-log scale, with a $1/N$ reference slope.

Table 3: All four MI estimators, head to head.

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	$\hat{I}_{\text{NSB}}$	I_{true} (ref.)
200	2.3511	2.1711	0.0578	0.5098	0.7329
500	1.7525	1.5035	0.2137	0.6833	0.7329
1000	1.5071	1.2711	0.3062	0.7369	0.7329
2000	1.2299	1.0367	0.4340	0.7416	0.7329
5000	1.0030	0.8842	0.5807	0.7678	0.7329
20000	0.8222	0.7716	0.6752	0.7411	0.7329

to know the size of the space being integrated over, not how much of it happened to be hit), then combine: $\hat{I}_{\text{NSB}} = \hat{S}_A^{\text{NSB}} + \hat{S}_B^{\text{NSB}} - \hat{S}_{AB}^{\text{NSB}}$.

NSB tracks I_{true} closely across the entire range, including at $N = 200$ where the naive estimator is off by roughly a factor of three and Miller–Madow has barely moved. This is the payoff of treating the whole problem as Bayesian inference over entropy rather than patching a biased point estimate after the fact. Figure 6 adds NSB to the convergence comparison: its residual curve sits visibly below the other three methods across most of the range, and in particular stays far more stable at small N – it doesn’t need $N \gg K$ to behave sensibly.

Practical notes for the real STAR dijet data.

- The alphabet size K (for each marginal, and $K_A \times K_B$ for the joint) needs to be fixed *before* looking at the data – pick a generous fixed upper bound on plausible hadron multiplicity per jet (e.g. from detector acceptance / kinematic limits), not the observed range, or a data-dependent bias is reintroduced through the back door.
- NSB is somewhat more expensive per call than MM or shuffle-subtraction (a 1D numerical

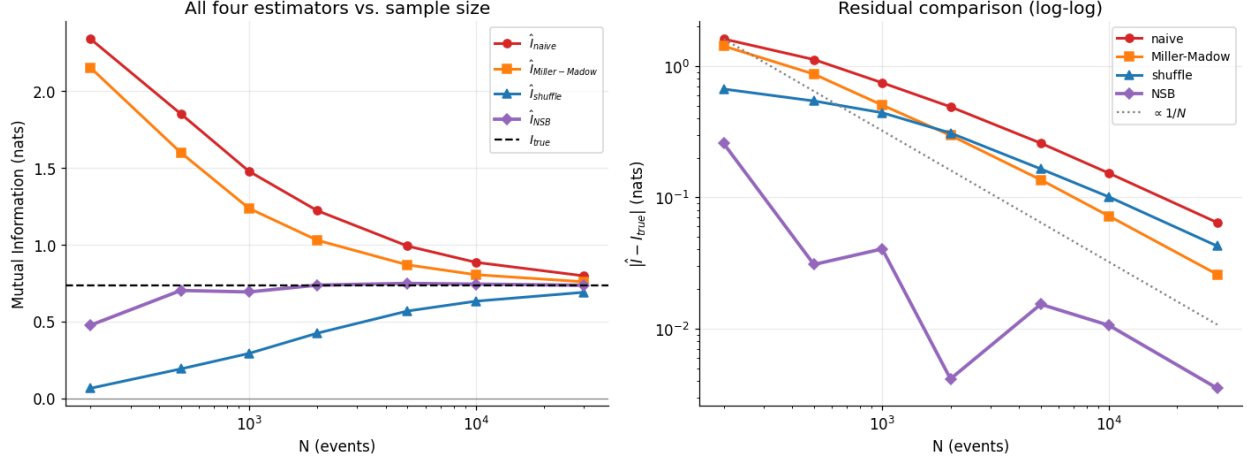


Figure 6: Left: all four estimators vs. sample size. Right: residual comparison on a log-log scale, with a $1/N$ reference slope.

integral instead of a closed-form correction or a handful of permutations), but it remains a fraction of a second per event sample – entirely practical for a real analysis, including a full bootstrap for uncertainty quantification (Sec. 7 works this out in detail).

7 Error estimation

Every number quoted so far needs an uncertainty attached to it, and the same machinery is used for that throughout this note – in the toy Monte Carlo study, in the PYTHIA 8 validation, and in every closure test. This section collects that machinery in one place, defines the parameters that control it (N_{boot} , $N_{repeats}$, $N_{shuffles}$) precisely, and states the general principle everything downstream follows.

7.1 The general principle: bootstrap the quantity itself

An uncertainty on any estimator here is obtained by **resampling events with replacement** from the actual sample being analyzed (a standard nonparametric bootstrap), recomputing the *entire* quantity of interest – $I(A:B)$ itself, or later the mixed-event difference $I_{same} - I_{mix}$ – on each resample, and taking the spread of that recomputed quantity across resamples as its standard error. The one rule that matters throughout this note: **a derived or composite quantity must be bootstrapped directly, together, from the same resamples** – never assembled by separately bootstrapping its pieces and combining their errors analytically afterward. The two are not interchangeable whenever the pieces are correlated (and they usually are, since they are computed from the same events): propagating component errors analytically either ignores that correlation or requires modeling it explicitly, while resampling the full composite quantity captures it automatically, at no extra conceptual cost. This distinction was not academic in developing this pipeline – an early version of the analysis bootstrapped one composite quantity correctly while computing a closely related one from a leftover analytic formula in the same function, giving visibly inconsistent, overly conservative error bars for the latter until the two were unified under one resampling procedure. The lesson generalizes: any pipeline computing multiple derived statistics from what is meant to be one bootstrap procedure should verify that *all* of them actually reuse the same resamples, not just that each individually looks plausible.

7.2 The three counters: N_{boot} , N_{repeats} , N_{shuffles}

Three distinct integers control the error-estimation machinery in this note, and they answer three different questions. Confusing them is an easy mistake, since all three superficially look like “how many times do we repeat something.”

- N_{boot} – **bootstrap resamples per point estimate.** For a single dataset (real or simulated), N_{boot} is the number of resample-with-replacement draws used to build *one* uncertainty σ on *one* point estimate. It controls how precisely that σ itself is known: the bootstrap standard error of a bootstrap standard deviation estimated from N_{boot} resamples scales roughly as $\sigma/\sqrt{2(N_{\text{boot}} - 1)}$, so $N_{\text{boot}} = 50$ pins down σ to within $\approx 10\%$, $N_{\text{boot}} = 200$ to $\approx 5\%$. Larger N_{boot} costs proportionally more compute per point estimate but does not change what is being measured, only how precisely its uncertainty is known.
- N_{repeats} – **independent fresh draws, closure tests only.** Where N_{boot} characterizes the uncertainty on *one* measurement, N_{repeats} is specific to the closure-test machinery (Sec. 9): the number of independent fresh samples drawn from the calibrated model, each analyzed completely independently (its own point estimate *and* its own N_{boot} -fold bootstrap σ), to build a *distribution* of pulls across repeats. This answers a meta-question – “is the uncertainty well-calibrated, on average, across many hypothetical realizations of this dataset” – that a single measurement’s own bootstrap can never answer about itself. N_{repeats} controls how trustworthy the reported pull *mean* and *standard deviation* themselves are: with $N_{\text{repeats}} = 30$, the pull standard deviation is itself only known to within roughly $\pm 13\%$ ($1/\sqrt{2 \times 29}$), so a closure test run with few repeats should be read as indicative, not precise, and the default of 30–100 used throughout this note is a practical compromise against the cost of also running a full N_{boot} -fold bootstrap *inside* every one of those repeats.
- N_{shuffles} – **label permutations, shuffle-correction only.** Specific to the shuffle-subtraction estimator (Sec. 5.2): the number of times the B -labels are randomly permuted to build that estimator’s own null distribution. It is internal to computing *one* shuffle-corrected point estimate and is unrelated to N_{boot} or N_{repeats} – in particular, it is irrelevant to NSB, which never shuffles anything.

In short: N_{shuffles} is consumed *inside* one shuffle estimate; N_{boot} resamples *are* one uncertainty estimate; N_{repeats} (closure tests only) repeats the entire measure-plus-its-own-uncertainty procedure to check whether that uncertainty behaves the way it claims to.

7.3 Where each is used in this note

- **Toy MC bias study** (Sec. 4): repeated draws at each sample size N are used directly to show the scatter of $\hat{I}_{\text{naive}}$ around its mean at that N (Fig. 2) – the same conceptual role as N_{repeats} above, though there the goal is visualizing the bias curve itself rather than a pull distribution against a bootstrap σ .
- **PYTHIA 8 validation** (Sec. 8): every reported point estimate and its error bar comes from a single N_{boot} -fold bootstrap on the actual generator sample, following Sec. 7.1 throughout – e.g. the multi-method comparison (Fig. 16) bootstraps each method’s $I(A:B)$ independently in every p_T bin.
- **Closure tests** (Sec. 9): N_{repeats} fresh draws from the calibrated model, each with its own N_{boot} -fold bootstrap, exactly as defined above.

- **Mixed-event subtraction** (Sec. 10): a further wrinkle on the same principle – since I_{same} and I_{mix} are computed from overlapping, correlated event samples, their *difference* is bootstrapped *jointly*: each of the N_{boot} resamples rebuilds both the same-event and the mixed-event pairs (re-matching partners within that resample) and recomputes the full difference, never bootstrapping the two terms separately and subtracting their errors afterward.

8 Validation on PYTHIA 8 Monte Carlo simulation

The toy model of Secs. 4–6 demonstrates that the estimator pipeline behaves correctly on data with a known, simple generative structure. The natural next step is to check the same pipeline on a full QCD event generator, where hadron multiplicities arise from parton showering, hadronization, and (optionally) the underlying event – none of which was built into the toy model by hand. We generated back-to-back charged-jet dijets with PYTHIA 8 (Monash 2013 tune, anti- k_T jets, $R = 0.4$, charged final-state particles only, matching a STAR-TPC-like charged-jet definition) and applied the identical naive/ Miller–Madow/shuffle/NSB pipeline, unmodified, to the generator output.

This particular validation run used LHC-like kinematics ($\sqrt{s} = 14$ TeV rather than the $\sqrt{s} = 200$ GeV RHIC energy that motivates the eventual STAR measurement) specifically *because* of its much wider accessible jet- p_T range: as discussed below (Sec. 8.2), the multiplicity-vs- p_T correlation predicted by QCD is intrinsically a slow, logarithmic effect, and a wide dynamic range makes it far easier to confirm the pipeline is tracking real physics rather than an artifact of a narrow kinematic window. The 200 GeV run for the actual STAR measurement is a separate, ongoing exercise (see Sec. 11); everything in this section is a *methods validation*, not a physics result.

8.1 Multiplicity and p_T distributions

Figure 7 shows the charged-multiplicity distributions for the leading and subleading jets, overlaid with a method-of-moments Negative-Binomial fit (as in Sec. 4). Both jets show genuine overdispersion relative to Poisson (var/mean = 1.55 for the leading jet, 1.29 for the subleading jet), qualitatively consistent with – though not identical in shape to – the NBD-like multiplicity distributions assumed in the toy model. This is a useful cross-check in its own right: the toy model’s central structural assumption (NBD-shaped marginals) is not merely a convenient fiction, but a reasonable approximation to what a full QCD generator actually produces.

Figure 8 shows the corresponding joint distribution: 36.9% of the 53×44 grid is occupied ($K_{AB} = 861$) at $N \approx 1.9 \times 10^5$ events – a moderately, not severely, undersampled regime, comparable to the middle of the range explored in the toy study. Figure 9 shows the (steeply falling, as expected for a QCD jet spectrum) leading- and subleading-jet p_T distributions, and Fig. 10 their joint distribution – the latter makes visually clear why quantile (rather than fixed-width) p_T binning is necessary for the differential measurements below: the bulk of events sit in a narrow low- p_T band, with a long, sparse tail.

8.2 The multiplicity- p_T correlation

A basic physical expectation – confirmed experimentally (e.g. ALICE, arXiv:2311.13322: “the mean charged-particle multiplicity inside the leading jet cone rises monotonically with increasing jet p_T ”) – is that higher- p_T jets should carry higher charged multiplicity, via additional phase space for parton radiation before hadronization. This growth is predicted by QCD coherent-branching (MLLA) arguments to be slow and roughly *logarithmic* in p_T , not linear, which matters directly

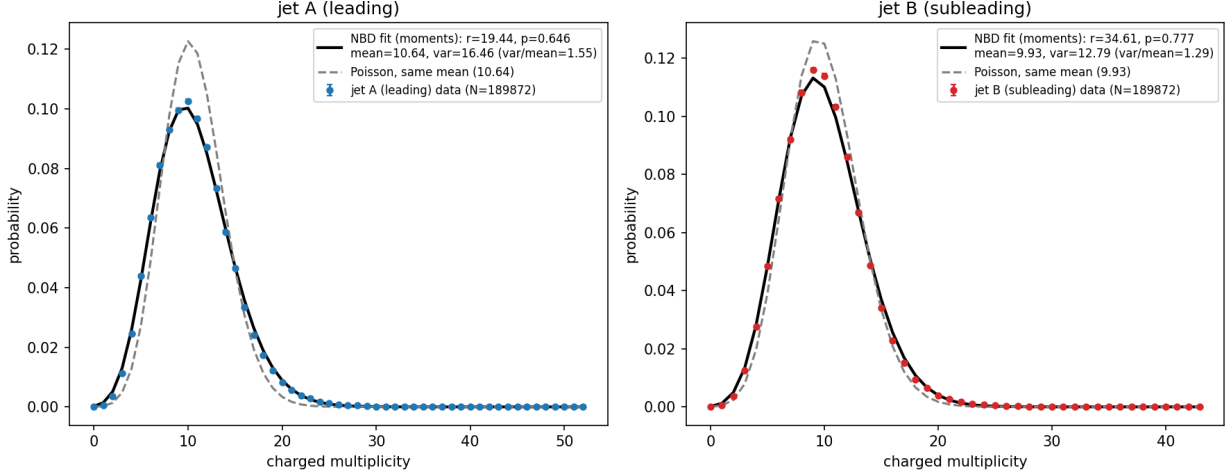


Figure 7: PYTHIA 8: charged-multiplicity distributions for the leading (left) and subleading (right) jet, with NBD and same-mean Poisson overlay (cf. the toy-model version, Sec. 4).

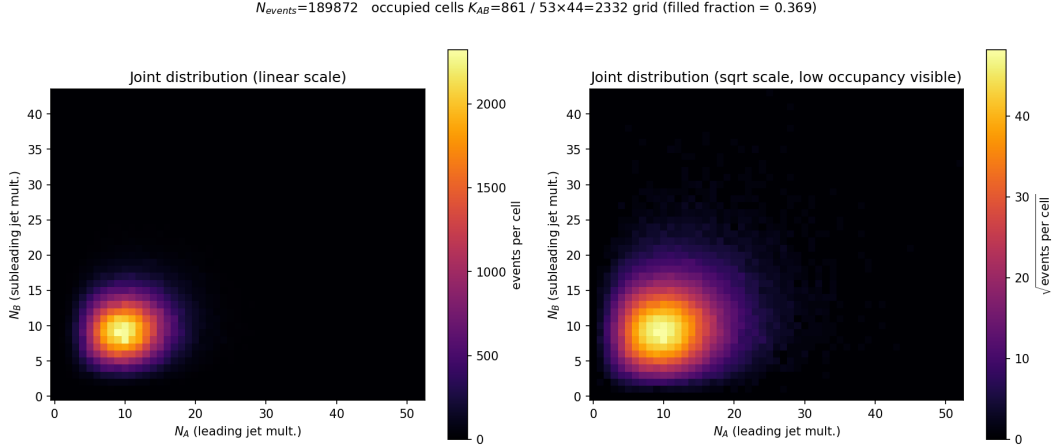


Figure 8: PYTHIA 8: joint charged-multiplicity distribution, linear and sqrt color scale (cf. Fig. 1).

for how easy the effect is to see over a limited kinematic range (as it was, initially, in an earlier 200 GeV validation attempt with a narrower accessible p_T window).

Figure 11 shows the raw, unbinned-by-any-entropy- machinery profile of mean charged multiplicity vs. jet p_T , together with a weighted fit of the MLLA-motivated form $\langle N \rangle = a + b \ln p_T$. The fitted slope is highly significant for both jets ($b = 3.77 \pm 0.05$, 83σ , leading jet; $b = 4.71 \pm 0.07$, 66σ , subleading jet) – direct, quantitative confirmation that PYTHIA 8 reproduces the expected QCD multiplicity-growth pattern, and that the analysis pipeline correctly recovers it from generator output with no prior assumption beyond bin means and standard errors.

8.3 Finite-sample bias, reproduced on generator output

Figure 12 repeats the toy study’s core diagnostic (Sec. 4) directly on subsamples of the real PYTHIA 8 output: the naive $\hat{I}$ falls from ≈ 0.66 nats at $N = 200$ down toward the full-sample NSB value (≈ 0.009 nats) as N grows to $\approx 1.9 \times 10^5$ – the same qualitative bias curve seen throughout

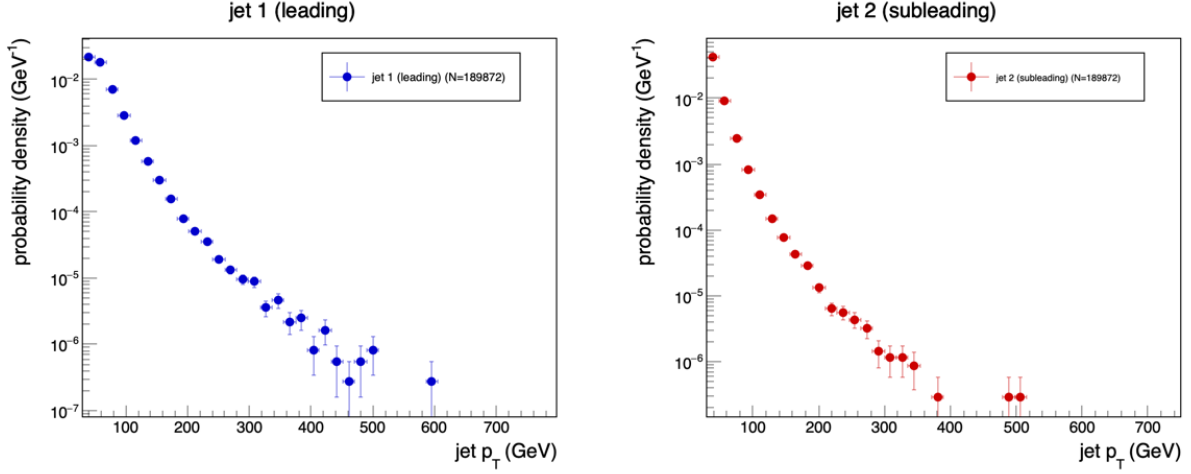


Figure 9: PYTHIA 8: leading- and subleading-jet p_T distributions (log-scale y -axis, to show the falling spectrum's tail).

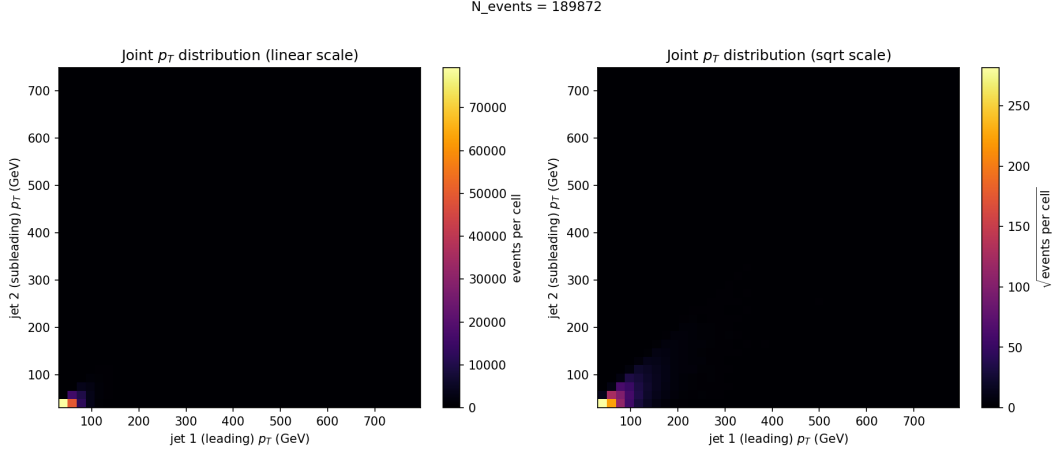


Figure 10: PYTHIA 8: joint leading/subleading jet p_T distribution, linear and sqrt color scale.

Sec. 4, now confirmed on a dataset the toy model was never built to resemble. This is a meaningful cross-check: the bias mechanism identified analytically (Sec. 3) and demonstrated on a synthetic Gamma-Poisson model is not an artifact of that specific generative structure – it appears with the same shape on genuine QCD Monte Carlo output.

8.4 Point estimates on the full sample

Figure 13 shows all four estimators on the full sample ($N \approx 1.9 \times 10^5$): naive (0.0112 ± 0.0004), Miller–Madow (0.0092 ± 0.0004), shuffle-corrected (0.0084 ± 0.0003), and NSB (0.0085 ± 0.0004 nats) – naive sits visibly above the other three, as expected from the bias mechanism above, while Miller–Madow/shuffle/NSB agree closely with each other, consistent with all three having converged toward the true value at this statistics (matching the closure seen among independently-biased estimators at large N in the toy study, Sec. 4). This measurement is a validation of the estimator machinery, not a physics claim.

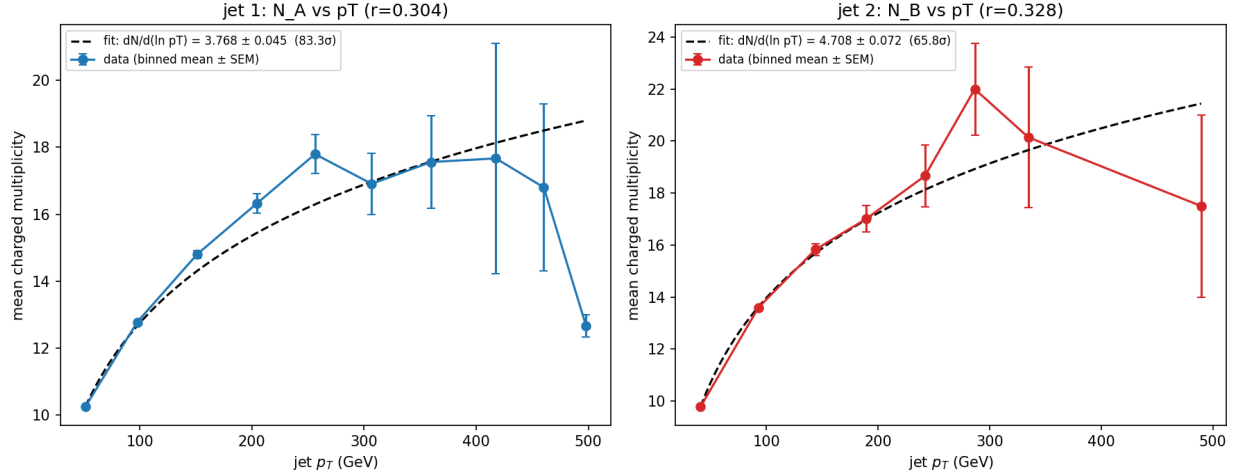


Figure 11: PYTHIA 8: raw mean charged multiplicity vs. jet p_T (binned mean $\pm$ SEM), with a weighted fit to $\langle N \rangle = a + b \ln p_T$. Both slopes are significant at $> 60\sigma$.

8.5 Differential measurement vs. jet p_T

Figures 14–15 show S_A, S_B, S_{AB} and $I(A:B)$ (NSB) in quantile bins of leading-jet p_T . All three entropies rise smoothly with p_T (Fig. 14), directly reflecting the multiplicity growth of Sec. 8.2. $I(A:B)$ (Fig. 15) also rises with p_T , modestly at low-to-mid p_T and more sharply in the highest (widest, most statistics-limited) bin.

8.6 Multi-method comparison vs. p_T

Figure 16 overlays naive/Miller–Madow/ shuffle/NSB for $I(A:B)$, each with bootstrap error bars computed by the same unified procedure (Sec. 7). The bias hierarchy naive $>$ Miller–Madow $\gtrsim$ NSB $\approx$ shuffle seen on the full sample (Fig. 13) persists across p_T bins, most visibly in the highest- p_T (lowest-statistics) bin, where naive’s upward bias is largest – exactly the pattern the toy study predicts for a bin with fewer events and a wider effective alphabet.

9 Closure test: calibrating estimator uncertainty on realistic data

The toy study (Secs. 4–6) validates the estimator suite against a *known* true MI, by construction. Real PYTHIA 8 output, and eventually real STAR/ALICE data, has no such known truth to validate against directly. This section describes a closure test that closes that gap as far as is practically possible: calibrate a synthetic model *to resemble the real data*, use that model to generate a pseudo-truth precise enough to treat as exact, and check whether the estimator pipeline recovers it under closure – i.e. under exactly the statistical conditions (sample size, alphabet, correlation strength) of the real measurement.

9.1 Method

1. **Calibrate.** Fit the toy study’s Gamma-Poisson shared-latent model (Sec. 4: $\Lambda \sim \text{Gamma}(k, \theta)$, $N_A|\Lambda \sim \text{Poisson}(\Lambda)$, $N_B|\Lambda \sim \text{Poisson}(c\Lambda)$) to the *real* data’s five leading moments – $\langle N_A \rangle$, $\text{Var}(N_A)$, $\langle N_B \rangle$, $\text{Var}(N_B)$, $\text{Cov}(N_A, N_B)$ – via least squares over (k, θ, c) . This is a genuine, honest constraint: a 3-parameter model

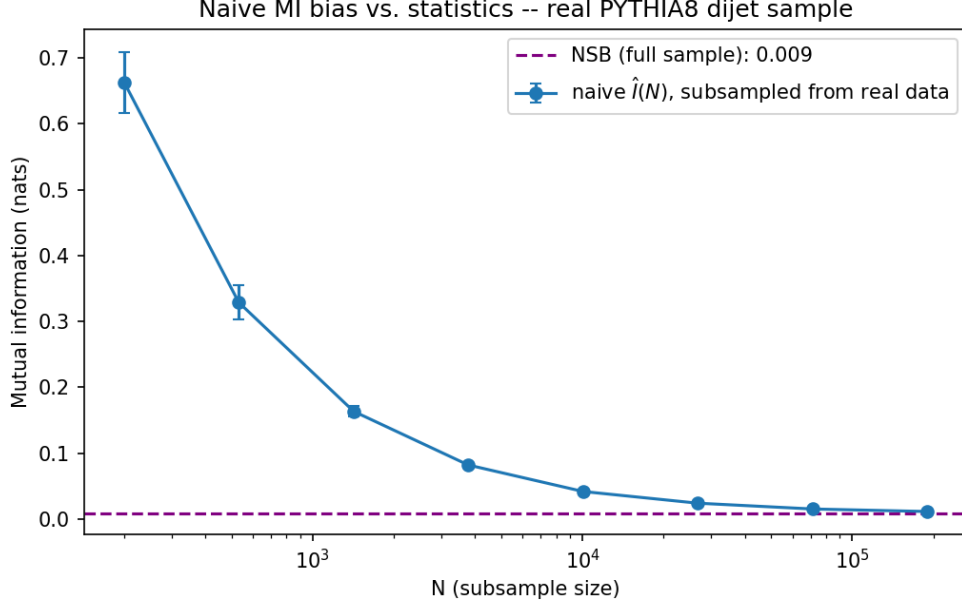


Figure 12: PYTHIA 8: naive $\hat{I}$ vs. subsample size N , real generator output (cf. the toy-model version, Fig. 2).

cannot exactly match 5 targets in general, and the achieved fit quality is reported explicitly rather than assumed.

2. **Establish a pseudo-truth.** Draw a very large sample (typically 10^6 – 5×10^6 events) from the *fitted* model. At this scale the naive estimator’s own bias ($\mathcal{O}(1/N)$, Sec. 3) is negligible, so $I(A:B)$ computed on this sample is treated as numerically exact (I_{true}) – calibrated to resemble the real measurement’s shape and correlation strength, not an arbitrary toy scenario.
3. **Closure.** Draw N_{repeats} independent fresh samples from the fitted model, each at the real dataset’s actual event count N (and using the real analysis’s alphabet), and run the full naive/Miller–Madow/shuffle/NSB pipeline on each.
4. **Check bias and calibration.** For NSB specifically, form the *pull* for each repeat, $z = (\hat{I}_{\text{NSB}} - I_{\text{true}})/\sigma_{\text{NSB}}$, using that repeat’s own bootstrap σ_{NSB} (Sec. 7). If the estimator is unbiased and its uncertainty well-calibrated, the pull distribution across repeats should be standard normal: mean 0, standard deviation 1, with $\sim 68\%$ of repeats within $\pm 1\sigma$ and $\sim 95\%$ within $\pm 2\sigma$.

This is purely diagnostic throughout – no bias or coverage correction is applied to any reported number anywhere in this note; the closure test’s role is to establish *whether* the raw bootstrap uncertainty can be trusted at face value for a given $(N, \text{alphabet})$, not to patch it.

9.2 Whole-sample results

Applying this to the full 14 TeV validation sample (Fig. 17) gives two consistent findings:

- **A mean pull of approximately -1 :** NSB systematically under-estimates $I(A:B)$ by

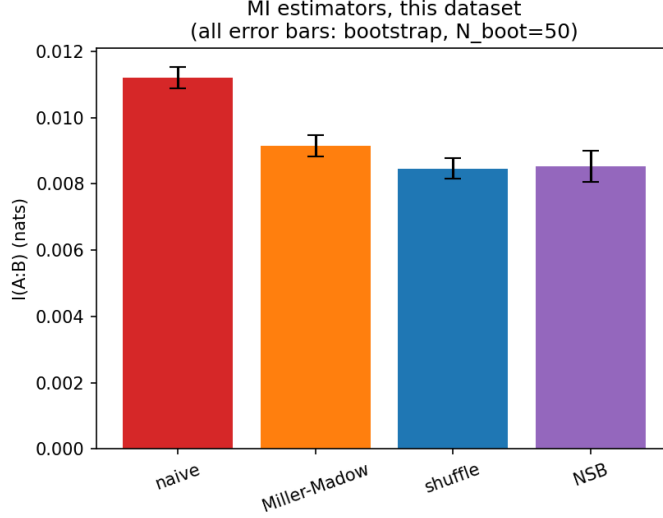


Figure 13: PYTHIA 8: all four MI estimators on the full sample, bootstrap error bars.

roughly one of its own reported bootstrap- σ 's, on average, at this sample's statistics. This is a real, non-negligible bias, though modest in scale.

- **A pull standard deviation of approximately 1.5–1.7:** the *actual* spread of $(\hat{I} - I_{\text{true}})$ across independent repeats is 50–70% *larger* than what the reported bootstrap σ claims. In other words, $\text{NSB} \pm \text{bootstrap } \sigma$ under-covers – the quoted error bars are too tight. A significance computed naively from the raw bootstrap σ at this $(N, \text{alphabet})$ therefore overstates the true significance by roughly this same factor (e.g. a naive 3σ claim would correspond to a true significance closer to $3/1.7 \approx 1.8\sigma$).

The pull distribution is also visibly non-Gaussian: skewed, with a heavier negative tail extending to several σ beyond what a standard normal would predict, rather than a uniformly-widened but still-symmetric Gaussian. This matters practically: a single multiplicative correction to σ (calibrating only the second moment) would not fully correct the shape of the tail, which is the part most relevant to a discovery-level claim specifically. Any downstream use of these bootstrap uncertainties for a significance statement should treat this as a known, quantified limitation rather than an assumption to be taken on faith – exactly the purpose of running this closure test before, not after, a real measurement.

9.3 p_T -differential closure test

The results of Sec. 9.2 calibrate NSB's uncertainty for a dataset shaped like the *whole* sample – but both the underlying multiplicity/correlation structure and the per-bin event count change with p_T (Sec. 8.2), so whole-sample calibration does not automatically transfer to any individual p_T bin, particularly the highest- p_T , lowest-statistics bins where a genuine signal would be most interesting and least well-measured. A p_T -differential extension of the closure test has been implemented: for each p_T bin independently, fit a bin-specific Gamma-Poisson model to that bin's own events (not the whole-sample fit), generate a bin-specific pseudo-truth, and run the closure procedure at that bin's actual event count – directly answering “is NSB's uncertainty trustworthy in *this* bin” rather than assuming the whole-sample answer applies uniformly.

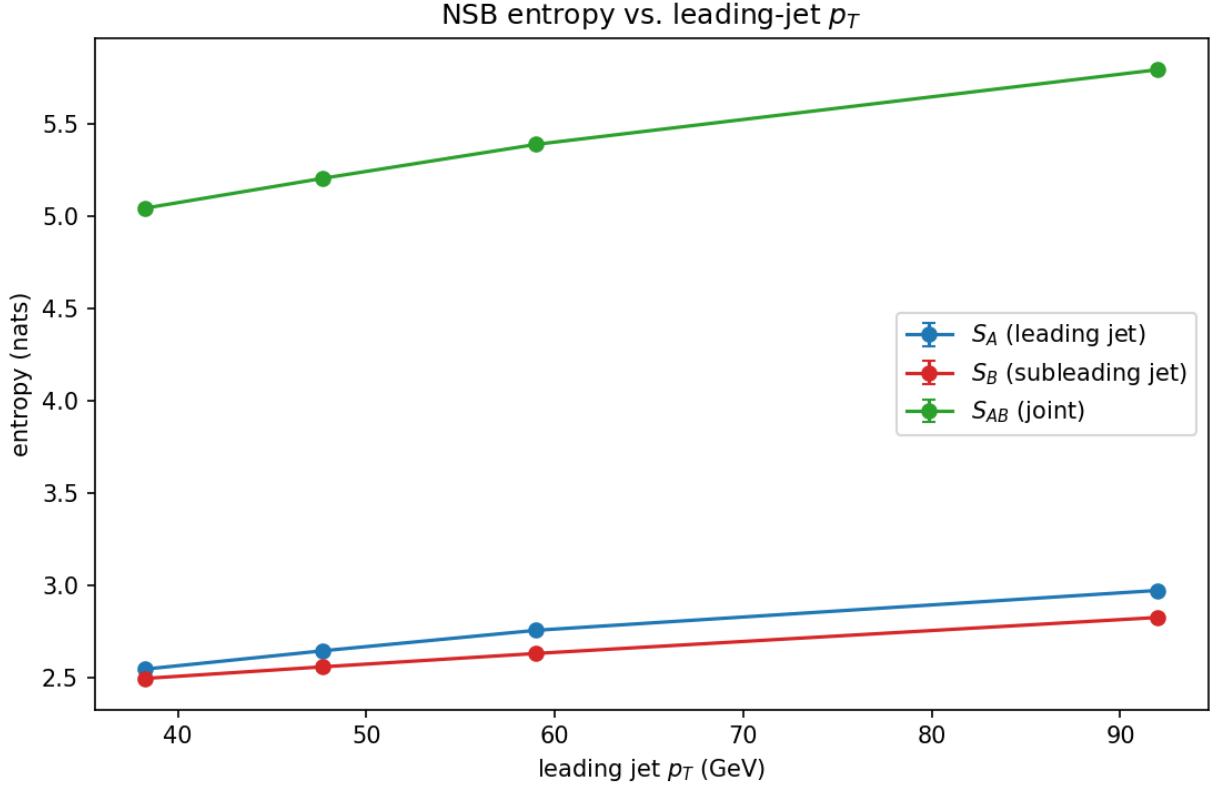


Figure 14: PYTHIA 8: S_A, S_B, S_{AB} (NSB) vs. leading-jet p_T .

Figure 18 shows the result, in the same four p_T bins used throughout Sec. 8.2 (p_T centers $\approx 38, 48, 59, 92$ GeV). The pattern is notably different from, and in one respect more encouraging than, the whole-sample result of Sec. 9.2:

- **Pull means** are small and negative throughout (≈ -0.12 to -0.17 in the three lower- p_T bins, shrinking to ≈ -0.04 in the highest- p_T bin) – markedly smaller in magnitude than the whole-sample ≈ -1 found in Sec. 9.2.
- **Pull standard deviations** rise from ≈ 0.58 (lowest- p_T bin, i.e. *over-covering* – the bootstrap σ is if anything too conservative there) through ≈ 0.80 and ≈ 0.95 to ≈ 0.92 in the highest- p_T bin – staying at or below 1 throughout, in contrast to the whole-sample 1.5–1.7 under-covering.

This bin-by-bin result being better-calibrated than the whole-sample one is itself a meaningful, self-consistent finding, not merely a nicer number: the whole-sample Gamma-Poisson fit has to compromise across a range of p_T -dependent correlation structures at once (Sec. 8.2), whereas each bin-specific fit only has to describe its own, comparatively homogeneous sub-population. A better-calibrated per-bin result is exactly what that design difference predicts, and is evidence the whole-sample closure test’s miscalibration (Sec. 9.2) is at least partly a model-fit-quality effect specific to forcing one 3-parameter model to span the full p_T range, rather than a property of NSB’s bootstrap uncertainty in general. That said, this is a single validation dataset at one set of statistics; whether this pattern holds at STAR’s actual 200 GeV kinematics and statistics (Sec. 11) is not yet established and should not be assumed.

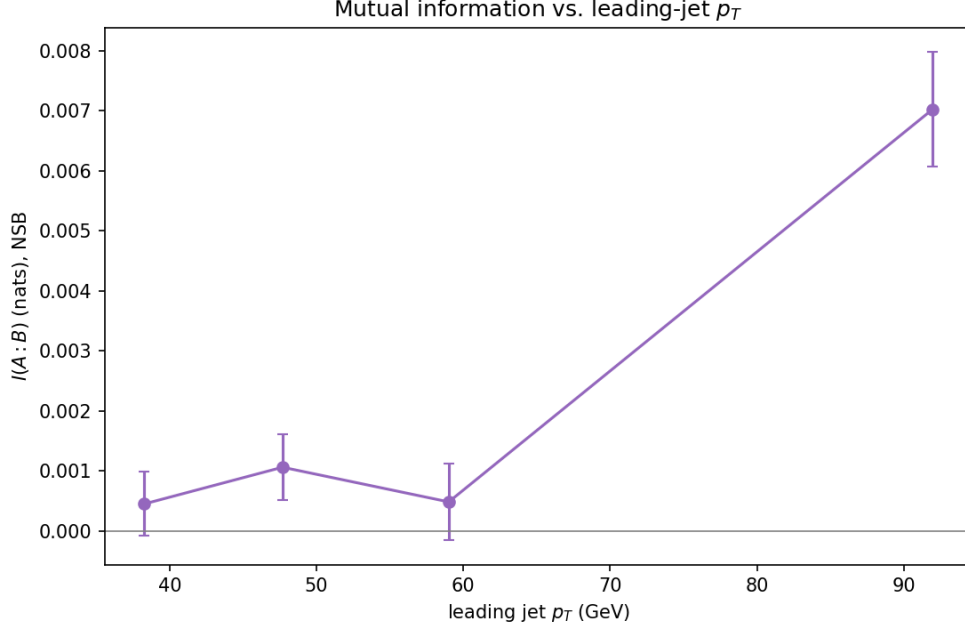


Figure 15: PYTHIA 8: $I(A:B)$ (NSB) vs. leading-jet p_T .

Even with this more encouraging per-bin picture, no bin here reaches pull std = 1 exactly, and the lowest- p_T bin’s over-covering (≈ 0.58) is itself a $\sim 40\%$ deviation from ideal calibration, in the conservative direction. The practical recommendation of Sec. 9.2 therefore still stands as the safe default – treat raw bootstrap σ with caution and consult a closure test at the actual bin/statistics in use – with this section’s result showing that the correction, at least for this validation sample, tends to be smaller and unidirectional (over-covering, not under-covering) per bin rather than the uniform large under-coverage seen whole-sample.

10 A distinct systematic: trivial kinematic MI, and its removal by mixed-event subtraction

Everything so far corrects *estimator* bias: the gap between a finite-sample MI estimate and the true MI of the sample’s generating distribution. This section addresses a different problem, upstream of estimation entirely – a sense in which the *true* MI being estimated is itself partly an artifact of the measurement’s finite p_T binning, not a real dynamical correlation.

10.1 The problem

Both jets’ multiplicities depend on their own p_T (Sec. 8.2), and $p_{T,1}$, $p_{T,2}$ are themselves correlated through momentum balance. This is a common-cause structure: even if N_A and N_B were completely independent *conditional on* the event kinematics, integrating over any p_T window of finite width induces real, nonzero $I(A:B)$ – not estimator bias, genuine correlation, but kinematically trivial and present in a purely classical generator with no dynamical jet–jet correlation at all. This does not vanish for any finite bin width, only shrinks as the bin narrows; the quantity one actually wants is closer to the conditional mutual information $I(N_A:N_B \mid p_T)$, which finite-width binning only approximates from above. This matters specifically for interpreting the *magnitude* of a measured

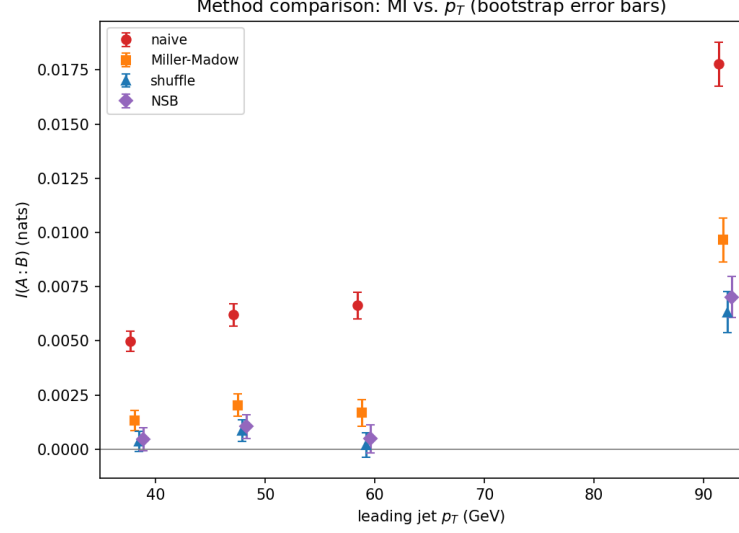


Figure 16: PYTHIA 8: method comparison for $I(A:B)$ vs. leading-jet p_T , bootstrap error bars throughout.

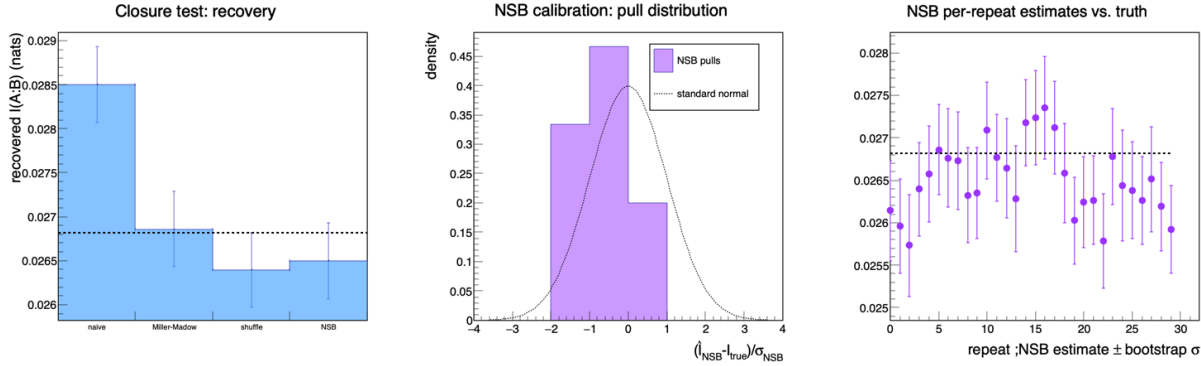


Figure 17: Whole-sample closure test, $I(A:B)$: recovery across methods (left), NSB pull distribution vs. standard normal (center), per-repeat NSB estimates vs. the fitted-model pseudo-truth (right).

$I(A:B)$ as evidence of dynamical (as opposed to purely kinematic) correlation, and does not depend on any witness-style bound – it is a statement about $I(A:B)$ itself.

10.2 Mixed-event subtraction

For each event, jet A’s multiplicity and kinematics are kept as measured; jet B’s multiplicity is replaced with that of a *different* event’s jet B, selected by k -nearest-neighbor matching in $p_{T,2}$ (default $k = 10$). Mixed pairs then reproduce the same kinematic structure as the same-event sample, but with N_B drawn conditionally independent of N_A given (matched) kinematics – so the mixed sample’s own MI, I_{mix} , directly estimates the trivial kinematic contribution. The beyond-kinematics signal is

$$I_{\text{signal}} = I_{\text{same}}^{\text{corrected}} - I_{\text{mix}}^{\text{corrected}}, \quad (13)$$

with bias correction applied to *each term separately* – not to the already-subtracted difference, which would double-correct, since the leading naive bias $\approx (K_{AB} - K_A - K_B + 1)/(2N)$ largely

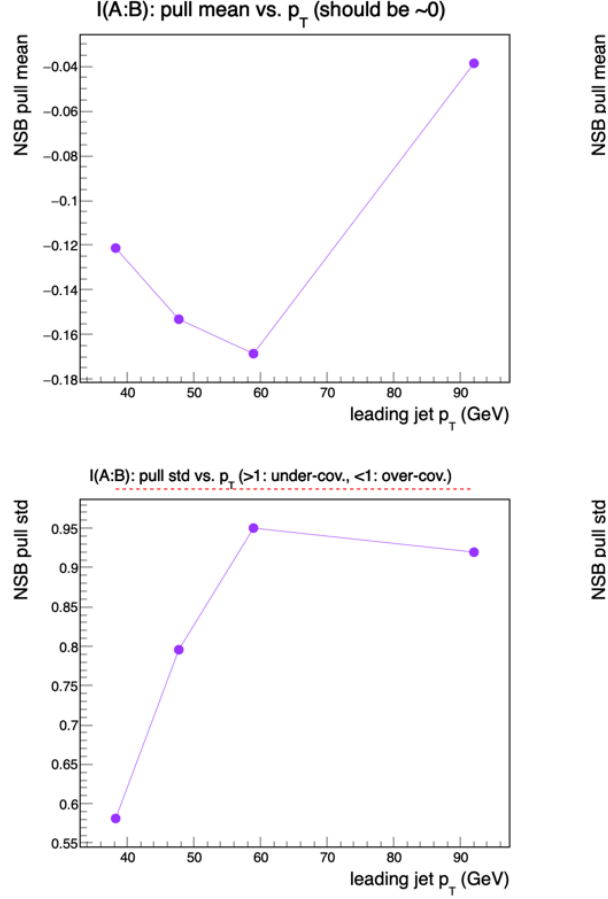


Figure 18: p_T -differential closure test, $I(A:B)$: NSB pull mean (top) and standard deviation (bottom) vs. leading-jet p_T . Red dashed line marks perfect calibration (pull std = 1).

cancels between the two terms already (same marginals and same N by construction). Uncertainties are obtained by a *joint* bootstrap of the full difference: each resample rebuilds both the same-event pairs and the mixed pairs (re-matching within the resample) and recomputes $I_{\text{same}} - I_{\text{mix}}$ for all four methods, never propagating the two terms' errors separately – the same principle as Sec. 7.1, and more important here since the two terms share events and are strongly correlated. The matching resolution k is itself a systematic: any residual within-window p_T variation remains unsubtracted trivial MI, and varying k and checking stability of I_{signal} is the natural check.

10.3 Relation to shuffle-subtraction

The two corrections are not redundant, though they are the same operation at different resolutions. Shuffling permutes N_B labels *within the analysis sample*, which realizes the product of that sample's own marginals – exactly what mixed-event pairing gives in the limit of no kinematic matching at all ($k \rightarrow$ the full sample). Formally: *shuffle-subtraction is mixed-event subtraction with matching resolution equal to the analysis bin*. What differs is the null each realizes, and so what each subtraction removes: shuffling's null has true MI exactly zero, so it corrects only finite-sample estimator bias and deliberately leaves all real correlation – including the trivial kinematic part – intact. Fine-matched mixing's null has nonzero true MI (the trivial kinematic part), so it removes

a genuine physical contribution. The two are complementary, not alternatives: bias correction (via shuffle, Miller–Madow, or NSB) is applied to each of I_{same} and I_{mix} individually, and the kinematic subtraction is taken between the two corrected numbers.

10.4 Validation: bias near-cancellation and closure test

Figure 19 first validates the construction on a purpose-built synthetic dataset with $N_A \perp N_B \mid p_T$ exactly by construction (a shared momentum-balance kinematic correlation with no dynamical jet–jet correlation added) – a case where I_{signal} should recover zero. The raw same-event MI is entirely trivial (≈ 0.034 – 0.039 nats depending on method) yet the mixed-event difference is consistent with zero for every one of naive, Miller–Madow, shuffle, and NSB. The near-cancellation predicted above is directly visible: the *raw* naive same-event and mixed-event estimates individually carry substantial bias (0.0386 vs. a true value near zero), but naive-minus-naive (-0.00132) lands almost on top of NSB-minus-NSB (-0.00140) – the bias has canceled between the two terms to within the bootstrap uncertainty, before any explicit correction was applied at all.

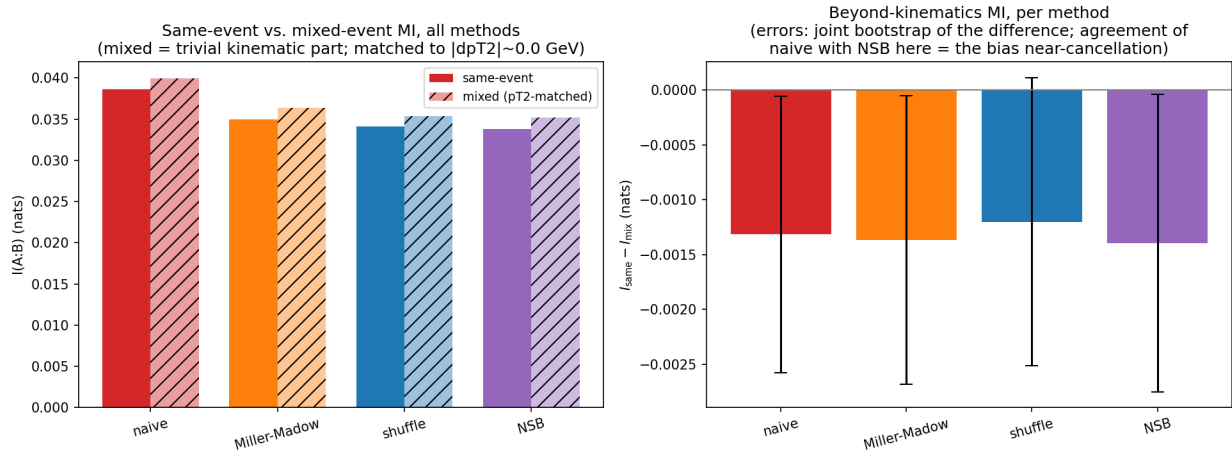


Figure 19: Mixed-event subtraction QA on a synthetic dataset with $N_A \perp N_B \mid p_T$ by construction. Left: same-event vs. $p_{T,2}$ -matched mixed-event $I(A:B)$, all four methods – both are dominated by the trivial kinematic contribution. Right: the per-method difference with jointly-bootstrapped errors, consistent with zero throughout; naive-minus-naive agreeing with NSB-minus-NSB despite the raw naive terms’ visible individual bias is the near-cancellation described in the text.

This motivated a dedicated closure test for the subtraction itself, distinct from Sec. 9’s closure tests for the raw estimators. The model resamples $(p_{T,1}, p_{T,2})$ pairs directly from real data (reproducing the true kinematic correlation exactly, no spectrum model needed), fits $\lambda_j(p_T) = a_j + b_j \ln p_T$ to each jet’s real multiplicity– p_T profile, and injects a controllable *genuine* beyond-kinematics correlation via a shared latent $g \sim \text{Gamma}(1/s^2, s^2)$ multiplying both jets’ rates ($s = 0$: exactly conditionally independent, true signal exactly zero; $s > 0$: a known injected signal). Pseudo-truth is obtained from a large reference sample analyzed with the identical matched-mixing procedure, so the truth is defined for exactly the estimand the method targets.

Unlike the reduced-statistics smoke test reported in an earlier revision of this note, Figs. 20–21 show the closure test run at full statistics (30 repeats, 30-fold joint bootstrap per repeat) directly on the real 14 TeV validation sample’s own fitted kinematics and multiplicity profile – both scenarios required by the design (Sec. 10): the *null* ($s = 0$), the essential false-positive control before claiming

any beyond-kinematics signal on real data, and an injected-signal case ($s = 0.15$).

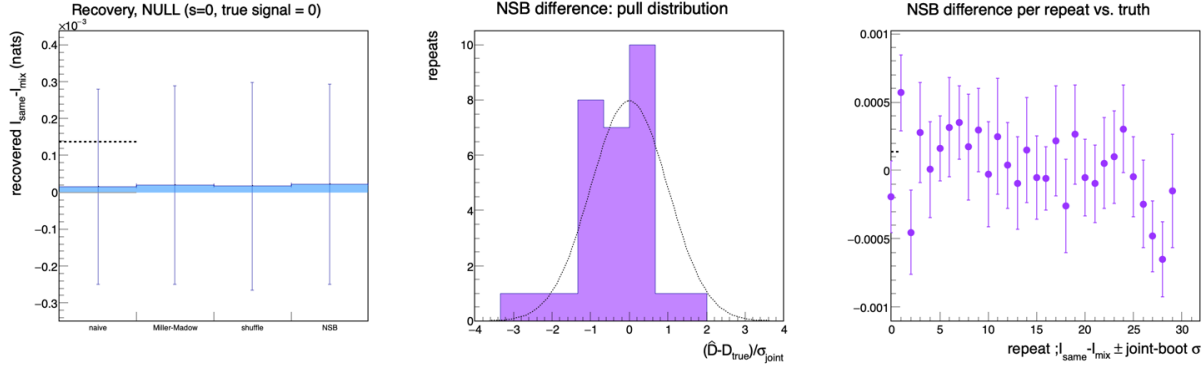


Figure 20: Closure test of the mixed-event subtraction on the real 14 TeV validation sample's fitted kinematics, null scenario ($s = 0$), full statistics (30 repeats). Left: recovered $I_{\text{same}} - I_{\text{mix}}$ per method vs. the (near-zero) pseudo-truth. Center: NSB-difference pull distribution against standard normal. Right: per-repeat NSB estimates with joint-bootstrap error bars vs. the true value.

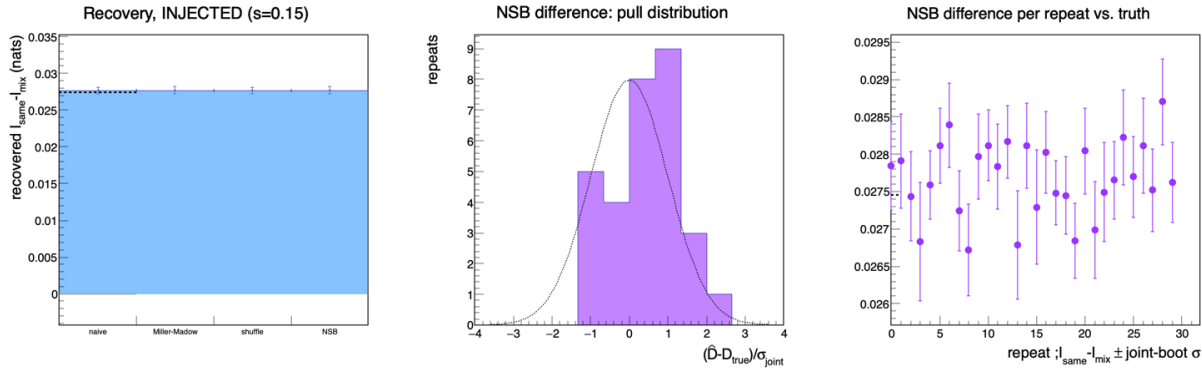


Figure 21: Same as Fig. 20, injected-signal scenario ($s = 0.15$, true difference ≈ 0.0275 nats).

In the null scenario the pseudo-truth itself is small but not exactly zero ($\approx 1.4 \times 10^{-4}$ nats) – expected, since it is estimated by the naive estimator on a large but finite reference sample and carries that estimator's own residual $\mathcal{O}(1/N)$ noise, not a flaw in the null construction. All four methods recover values consistent with this truth within their (comparatively large, since the true difference is itself tiny) joint-bootstrap uncertainties, and the NSB-difference pull distribution does not show the pronounced skew seen in the raw-estimator whole-sample closure tests (Sec. 9.2). In the injected-signal scenario, all four methods recover the true ≈ 0.0275 nats difference closely and consistently, again without the earlier tests' marked under-coverage. Taken together, the mixed-event *difference* appears better-calibrated at these statistics than the raw single-term estimators were – plausibly because the joint bootstrap directly captures the strong correlation between the same-event and mixed-event terms (they share events), which the raw single-quantity closure tests had no analogue of. This is an encouraging empirical finding rather than a proven general property, and should not be assumed to hold automatically at STAR's real kinematics and statistics without rerunning this same check there.

10.5 Application to the real validation sample

With the method validated, it was applied to the actual 14 TeV PYTHIA 8 validation sample used throughout Sec. 8 – answering the question Sec. 10’s design was motivated by: how much of the $I(A:B)$ measured elsewhere in this note is trivial kinematic correlation, versus a beyond-kinematics component.

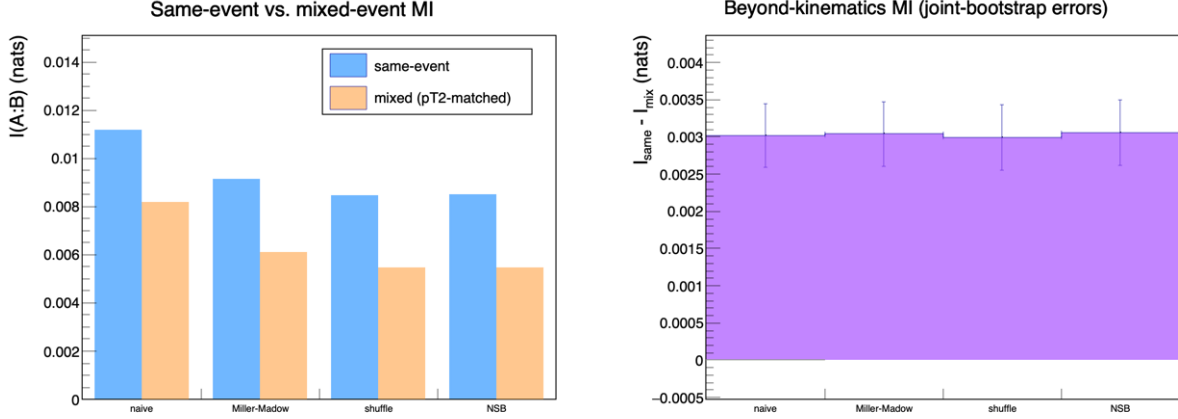


Figure 22: Mixed-event subtraction applied to the real 14 TeV validation sample (whole sample). Left: same-event vs. $p_{T,2}$ -matched mixed-event $I(A:B)$, all four methods. Right: the per-method difference with jointly-bootstrapped errors – a nonzero beyond-kinematics component, consistently recovered across all four estimators.

Figure 22 shows the whole-sample result. Same-event $I(A:B) \approx 0.0084$ – 0.0112 nats depending on method (naive highest, as expected from its known upward bias), dropping to ≈ 0.0055 – 0.0082 nats for the $p_{T,2}$ -matched mixed sample. The difference is ≈ 0.0030 nats, and – notably – essentially *the same* value across all four methods, with joint-bootstrap uncertainty $\approx \pm 0.0004$ – 0.0005 nats. Roughly 60–70% of the raw same-event MI is removed as trivial kinematic correlation (depending slightly on method); the remaining ≈ 0.003 nats is a genuine, nonzero beyond-kinematics signal, not an artifact of any one estimator’s bias.

Since PYTHIA 8 has no mechanism for quantum entanglement between separately hard-scattered jets, this remaining correlation must be classical in origin. The most likely candidate, anticipated already in Sec. 10’s design discussion, is shared underlying-event activity – multiple-parton interactions, beam-remnant particles, and color reconnection all act coherently on *both* jets within the same event, and matching mixed partners only on jet-B’s p_T does nothing to remove event-level (as opposed to jet-level) shared structure. This is consistent with, not a contradiction of, the subtraction’s design: the method isolates correlation beyond simple p_T -driven kinematics, and underlying-event effects are exactly the kind of genuine (if unglamorous) dynamical correlation that description was always going to include.

The p_T -differential breakdown (Fig. 23) is less clean, as expected given each bin carries roughly a quarter of the whole sample’s statistics: the two lower- p_T bins are consistent with zero, the third bin shows a small positive value (≈ 0.0009 – 0.0011 nats), and the highest- p_T bin shows the largest value (≈ 0.0015 – 0.0022 nats) but also by far the largest uncertainty. All four methods agree closely within each bin, which is reassuring for the subtraction’s internal consistency, but no individual bin here should be read as an independently significant detection – the whole-sample result in Fig. 22, with its higher statistics and tighter relative uncertainty, is the more trustworthy number at this

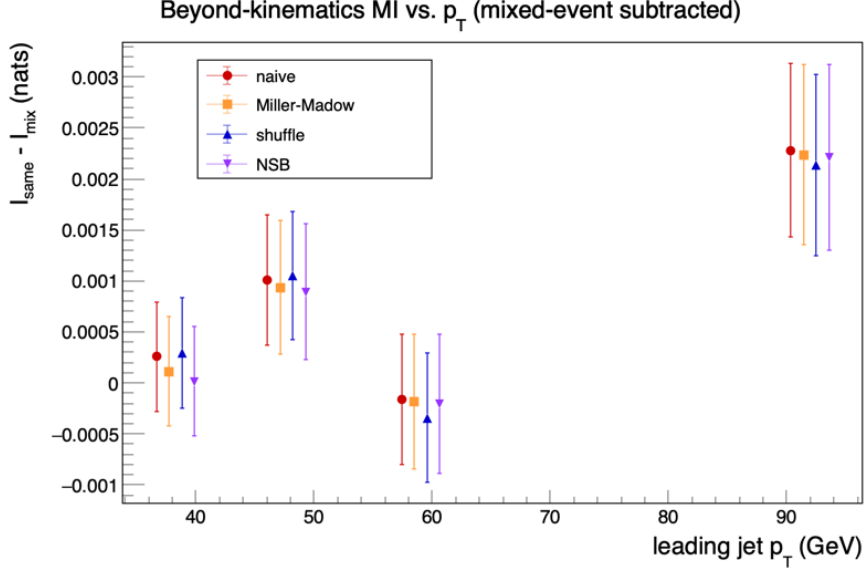


Figure 23: Beyond-kinematics MI vs. leading-jet p_T (mixed-event subtracted), real validation sample, all four methods.

stage. A closure test run specifically at each bin's statistics (the p_T -differential analogue of Sec. 9.3, not yet built for the mixed-event method) would be needed before quoting a per-bin significance with confidence.

10.6 Status and outlook

The subtraction machinery, its bias-cancellation property, a closure-test framework validating it at full statistics on the real sample's own kinematics, and an application to the real validation sample itself are now all complete (Figs. 19–23). What remains: a p_T -differential closure test for the mixed-event difference specifically (paralleling Sec. 9.3's extension of the raw-estimator closure test), needed before the per-bin breakdown in Fig. 23 can support a significance claim in any individual bin; and, longer-term, isolating the underlying-event contribution more directly (e.g. by matching on additional event activity variables, not just jet-B's p_T) if a cleaner separation of kinematic, underlying-event, and (eventually, on real detector data) any genuinely anomalous component is wanted.

11 Overall summary and next steps

This note works through one measurement – mutual information between two back-to-back jets' hadron multiplicities – end to end: a controlled toy study, validation on a full QCD event generator, a general error-estimation framework, closure tests that calibrate what the reported uncertainties are actually worth, and a further correction for a kinematic confound that has nothing to do with any of the above. This section summarizes what each stage established, in that order, and what remains before the identical pipeline can be applied to real STAR/ALICE data.

11.1 From the toy Monte Carlo study (Secs. 4–6)

1. **The naive plug-in MI estimator is always biased upward**, with leading term $\approx (K_{AB} - K_A - K_B + 1)/(2N)$. Coarser binning or fewer events both make it worse – comparing different STAR running periods, luminosities, or binning choices will report different central MI values from what should be the same physics, purely from this effect.
2. **Miller–Madow** (done correctly, using the observed K_{AB} , not the product $K_A K_B$) and **shuffle-subtraction** both substantially reduce this bias and stay well-behaved, but neither is exact at small N : Miller–Madow is a first-order truncation; shuffle-subtraction tends to systematically overcorrect when the true correlation is strong.
3. **NSB** is the recommended estimator for the real analysis: a Bayesian posterior-mean entropy built on a prior specifically constructed to be uninformative about entropy itself. It tracks the truth closely across the entire tested N range, including deep in the severely-undersampled regime where the other methods struggle, at a modest computational cost.

11.2 From the PYTHIA 8 validation (Sec. 8)

The same pipeline, unmodified, correctly recovers physically-expected behavior (NBD-like overdispersion, MLLA-consistent multiplicity growth with p_T , the same finite-sample bias shape) on realistic QCD Monte-Carlo output where no such structure was hand-built in – a meaningful, if indirect, validation that the pipeline is measuring real structure rather than an artifact of the toy model’s specific construction. An independent ROOT/C++ re-implementation cross-validates the Python results throughout.

11.3 From error estimation and closure testing (Secs. 7–9)

Sec. 7 fixes the general rule used everywhere in this note: bootstrap the full quantity of interest directly, never propagate separately-estimated component errors. A closure test calibrated to resemble the real PYTHIA 8 sample (Gamma-Poisson model fit to the real data’s marginals and correlation strength, huge reference sample for a pseudo-truth, repeated closure at the real dataset’s actual statistics) finds that NSB’s bootstrap uncertainty, at this sample’s (N , alphabet), is **under-covering by a factor of roughly 1.5–1.7**, with a mean bias of order -1σ and a pull distribution that is skewed rather than symmetrically widened. This is a quantitative, load-bearing result for any future significance claim: a naive 3σ excess computed from raw bootstrap σ at comparable statistics should be read as $\sim 1.8-2\sigma$ once this calibration factor is accounted for. No correction has been baked into the pipeline itself – this factor is reported here as a measured property of the estimator at these statistics, to be applied explicitly and cited whenever a significance is quoted, rather than silently folded into the numbers.

The p_T -differential extension of the closure test (Sec. 9.3) adds an important nuance: bin-by-bin, NSB’s pull standard deviations sit much closer to the ideal value of 1 ($\approx 0.58-0.95$) than the whole-sample result above, consistent with each bin’s own Gamma-Poisson fit describing a more homogeneous sub-population than the one compromise fit spanning the whole p_T range has to. This suggests the whole-sample under-coverage found in Sec. 9.2 is at least partly a consequence of fitting one model across a range of p_T -dependent structure, rather than an intrinsic property of NSB’s bootstrap procedure – but this is one validation dataset at one set of statistics, and the pattern should be re-checked, not assumed, at STAR’s actual kinematics.

11.4 From the kinematic-background correction (Sec. 10)

Sec. 10 addresses a separate concern from estimator calibration: any finite p_T window induces real but kinematically trivial MI through the momentum-balance correlation between the two jets' p_T 's. A mixed-event subtraction removes this by construction and exhibits a useful property found rather than assumed: the leading finite-sample bias very nearly cancels between the same-event and mixed-event terms, so the subtracted difference is close to unbiased even before explicit correction – confirmed both on a controlled synthetic dataset and by a full-statistics closure test run directly on the real sample's fitted kinematics, where the mixed-event difference showed better calibration than the raw single-term estimators of Sec. 9. Applied to the real 14 TeV validation sample, roughly 60–70% of the raw whole-sample $I(A:B)$ is trivial kinematic correlation; the remaining ≈ 0.003 nats, consistent across all four estimator methods, is a genuine beyond-kinematics signal most plausibly attributable to shared underlying-event activity (Sec. 10.5) rather than anything exotic – exactly the kind of confound this correction exists to separate out before any dynamical interpretation of a measured $I(A:B)$ magnitude is warranted. As discussed in Sec. 1, that interpretation question is bounded on both ends: any observed conditional-entropy violation could only ever be a statistical artifact (never directly measured physics), and genuine quantum discord is identically zero for any classical count distribution regardless. What this pipeline delivers is a rigorously measured $I(A:B)$, with its estimator bias removed, its uncertainty calibrated, and its trivial kinematic component separated out – the necessary, but not by itself sufficient, input to that larger question.

11.5 Remaining steps toward the actual STAR measurement

1. Repeat Sec. 8 at $\sqrt{s} = 200$ GeV with STAR-appropriate jet kinematics rather than the wide-dynamic-range 14 TeV validation kinematics used here.
2. Apply the detector-level acceptance and tracking-efficiency model (already implemented in the generator, not yet exercised in this note) and repeat the full analysis chain.
3. Rerun the whole-sample and p_T -differential closure tests (Secs. 9.2–9.3) at the actual statistics of (1)–(2), obtaining calibration factors specific to the real analysis rather than relying on the 14 TeV validation sample's factors as a proxy.
4. Rerun the mixed-event subtraction and its closure test (Sec. 10) at STAR kinematics and statistics – done so far only on the 14 TeV validation sample – including the still-missing p_T -differential closure test for the subtracted quantity, needed before any per-bin beyond-kinematics result can carry a trustworthy significance.
5. Once (1)–(4) are complete, apply the identical pipeline to real STAR data, with all applicable closure-test calibration factors applied explicitly before unblinding, so that any reported excess is quoted against a pre-calibrated, honestly-covered uncertainty rather than a raw bootstrap σ known already to under-cover.